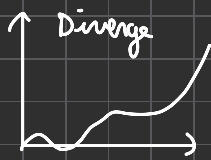
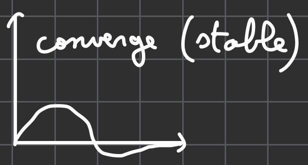
02) Stability in control

$$U(s) \longrightarrow \boxed{G(s)} \longrightarrow Y(s)$$

$$G(s) = \frac{a_m S^m + a_{m-1} S^{m-1} + \dots + a_0}{b_n S^n + b_{n-1} S^{n-1} + \dots + b_0}$$

To $y(t)$ not to diverge, when input $u(t)$ is applied, we need to determine the stability of system G.





Stability in State-Space Model.

$$\dot{x} = Ax + Bu \qquad \Longrightarrow \qquad \det(sI - A) = 0$$
$$y = Cx$$

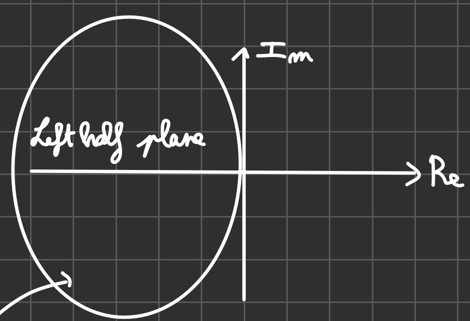
Characteristic equation

$\Downarrow$ State model to equivalent transfert function.

$$G(s) = C(sI - A)^{-1} B = \frac{a_m s^m + a_{m-1} s^{m-1} + \dots + a_0}{b_n s^n + b_{n-1} s^{n-1} + \dots + b_0}$$

Eigenvalue of A = Solutions in characteristic equation = Poles in open-loop system

If real part of eigenvalue of A are all negative, the system is asymptotically stable



if eigenvalues in here $\Leftrightarrow$ stable.

If real parts of eigenvalue of A are all negative, the system is asymptotically stable.

Routh's Stability Criterion.

Lyapunov Stability.

$V(x) > 0$ for all x and $V(0) = 0$

(Example. $V(x) = x^T P x$)

The time derivative $\dot{V} \leq 0$ (Non-increasing over time)

$\Rightarrow$ The function is named as "Lyapunov function"

Check stability of LTI system $\dot{x} = Ax$

Lyapunov Equation : $A^T P + PA + Q = 0$

If there exist $P > 0$, $Q \geq 0$, the system is asymptotically stable

($x^T P x$ is a Lyapunov funct)

$V = x^T P x$ is always positive.

Time derivative $\dot{V} = x^T (A^T P + PA) x = - x^T Q x$ holds and is non-increasing.